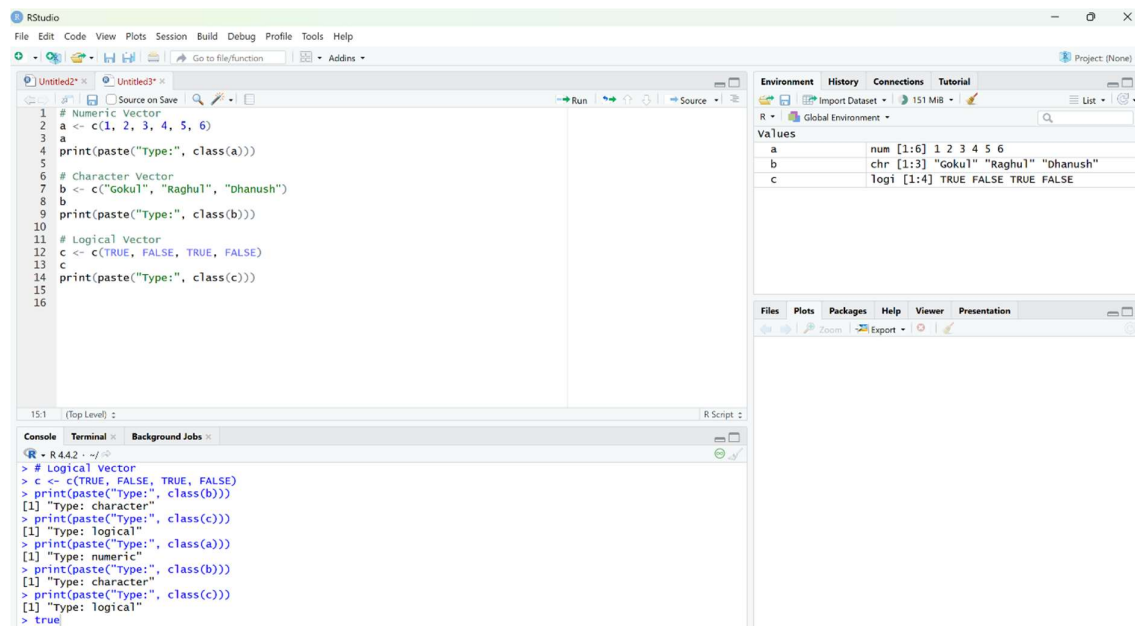


Experiment No : 1



The screenshot shows the RStudio interface. The source editor contains R code for creating and printing vectors. The console shows the output of the code. The environment pane on the right lists the objects created.

```
1 # Numeric Vector
2 a <- c(1, 2, 3, 4, 5, 6)
3 a
4 print(paste("Type:", class(a)))
5
6 # Character Vector
7 b <- c("Goku", "Raghu", "Dhanush")
8 b
9 print(paste("Type:", class(b)))
10
11 # Logical Vector
12 c <- c(TRUE, FALSE, TRUE, FALSE)
13 c
14 print(paste("Type:", class(c)))
15
16
```

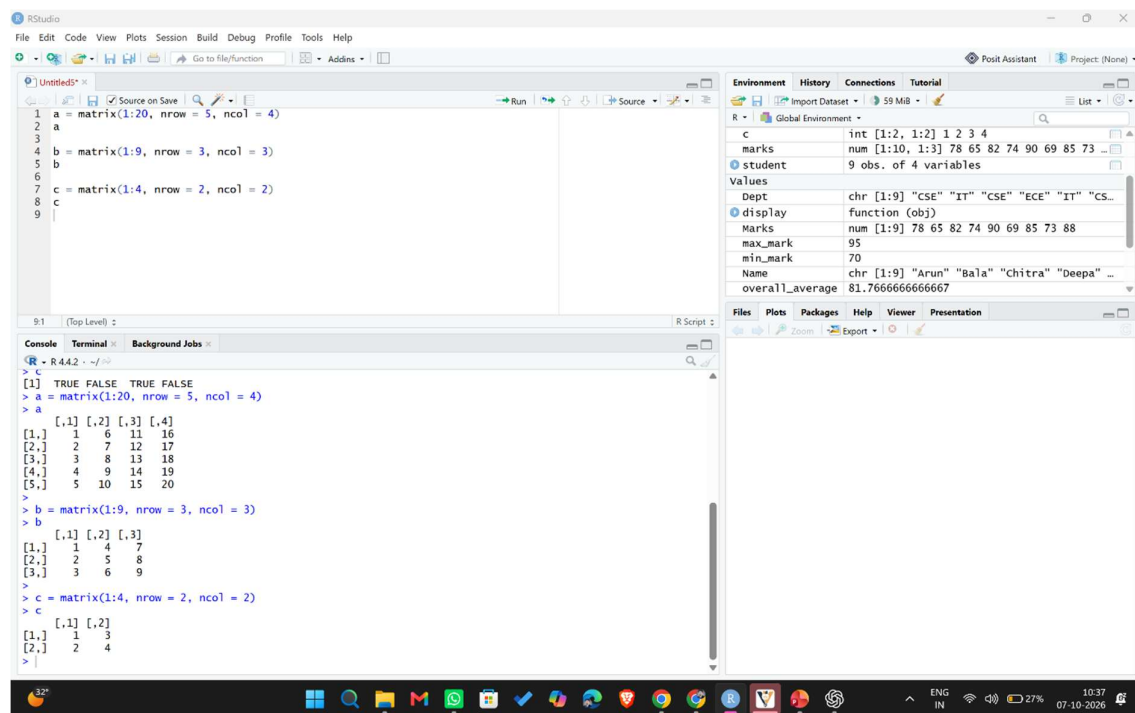
Console Output:

```
> # Logical Vector
> c <- c(TRUE, FALSE, TRUE, FALSE)
> print(paste("Type:", class(b)))
[1] "Type: character"
> print(paste("Type:", class(c)))
[1] "Type: logical"
> print(paste("Type:", class(a)))
[1] "Type: numeric"
> print(paste("Type:", class(b)))
[1] "Type: character"
> print(paste("Type:", class(c)))
[1] "Type: logical"
> true
```

Environment:

Object	Class	Values
a	num [1:6]	1 2 3 4 5 6
b	chr [1:3]	"Goku" "Raghu" "Dhanush"
c	logi [1:4]	TRUE FALSE TRUE FALSE

Experiment : 2



The screenshot shows the RStudio interface. The source editor contains R code for creating and printing matrices. The console shows the output of the code. The environment pane on the right lists the objects created.

```
1 a = matrix(1:20, nrow = 5, ncol = 4)
2 a
3
4 b = matrix(1:9, nrow = 3, ncol = 3)
5 b
6
7 c = matrix(1:4, nrow = 2, ncol = 2)
8 c
9
```

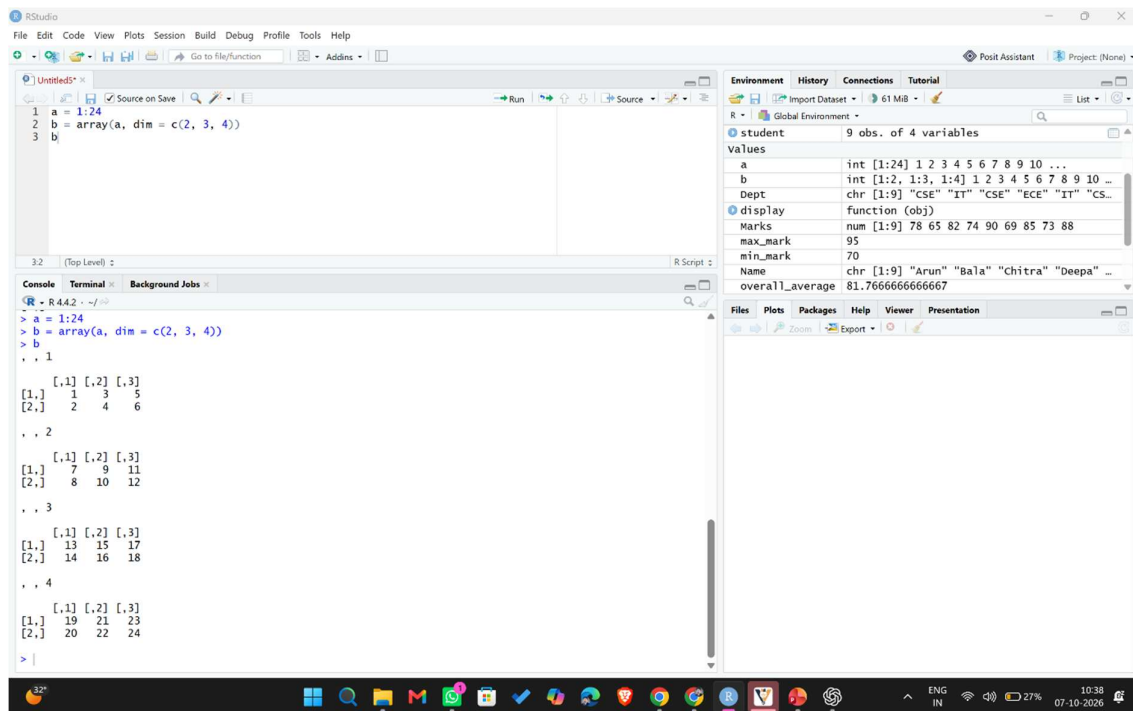
Console Output:

```
> c
[1] TRUE FALSE TRUE FALSE
> a = matrix(1:20, nrow = 5, ncol = 4)
> a
     [,1] [,2] [,3] [,4]
[1,]  1    6   11   16
[2,]  2    7   12   17
[3,]  3    8   13   18
[4,]  4    9   14   19
[5,]  5   10   15   20
> b = matrix(1:9, nrow = 3, ncol = 3)
> b
     [,1] [,2] [,3]
[1,]  1    4    7
[2,]  2    5    8
[3,]  3    6    9
> c = matrix(1:4, nrow = 2, ncol = 2)
> c
     [,1] [,2]
[1,]  1    3
[2,]  2    4
```

Environment:

Object	Class	Values
c	int [1:2, 1:2]	1 2 3 4
marks	num [1:10, 1:3]	78 65 82 74 90 69 85 73 ...
student	9 obs. of 4 variables	
Dept	chr [1:9]	"CSE" "IT" "CSE" "ECE" "IT" "CS..."
display	function (obj)	
Marks	num [1:9]	78 65 82 74 90 69 85 73 88
max_mark		95
min_mark		70
Name	chr [1:9]	"Arun" "Bala" "Chitra" "Deepa" ...
overall_average		81.7666666666667

Experiment : 3



The screenshot shows the RStudio interface for Experiment 3. The script editor contains the following code:

```
1 a = 1:24
2 b = array(a, dim = c(2, 3, 4))
3 b
```

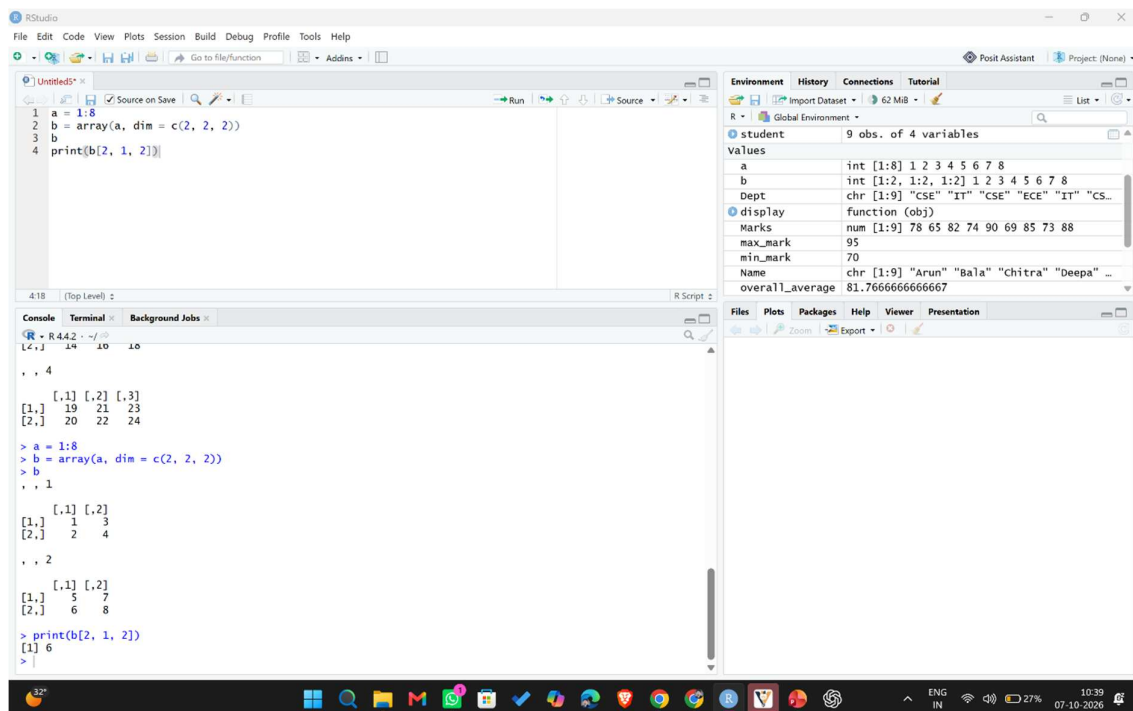
The console output shows the array structure and its values:

```
> a = 1:24
> b = array(a, dim = c(2, 3, 4))
> b
, , 1
[1,] 1 3 5
[2,] 2 4 6
, , 2
[1,] 7 9 11
[2,] 8 10 12
, , 3
[1,] 13 15 17
[2,] 14 16 18
, , 4
[1,] 19 21 23
[2,] 20 22 24
>
```

The Environment pane on the right shows the 'student' object with 9 observations and 4 variables:

Variable	Values
a	int [1:24] 1 2 3 4 5 6 7 8 9 10 ...
b	int [1:2, 1:3, 1:4] 1 2 3 4 5 6 7 8 9 10 ...
Dept	chr [1:9] "cse" "IT" "cse" "ECE" "IT" "CS_..."
display	function (obj)
Marks	num [1:9] 78 65 82 74 90 69 85 73 88
max_mark	95
min_mark	70
Name	chr [1:9] "Arun" "Bala" "Chitra" "Deepa" ...
overall_average	81.7666666666667

Experiment : 4



The screenshot shows the RStudio interface for Experiment 4. The script editor contains the following code:

```
1 a = 1:8
2 b = array(a, dim = c(2, 2, 2))
3 b
4 print(b[2, 1, 2])
```

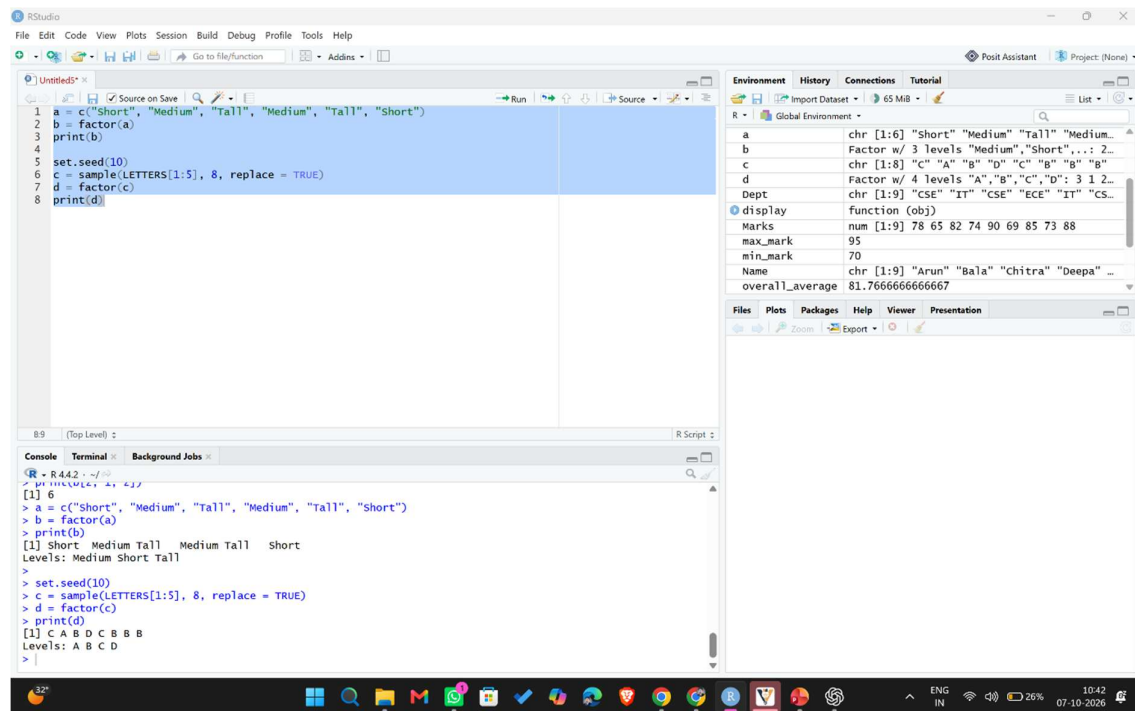
The console output shows the array structure and its values:

```
> a = 1:8
> b = array(a, dim = c(2, 2, 2))
> b
, , 1
[1,] 1 3
[2,] 2 4
, , 2
[1,] 5 7
[2,] 6 8
> print(b[2, 1, 2])
[1] 6
>
```

The Environment pane on the right shows the 'student' object with 9 observations and 4 variables:

Variable	Values
a	int [1:8] 1 2 3 4 5 6 7 8
b	int [1:2, 1:2, 1:2] 1 2 3 4 5 6 7 8
Dept	chr [1:9] "cse" "IT" "cse" "ECE" "IT" "CS_..."
display	function (obj)
Marks	num [1:9] 78 65 82 74 90 69 85 73 88
max_mark	95
min_mark	70
Name	chr [1:9] "Arun" "Bala" "Chitra" "Deepa" ...
overall_average	81.7666666666667

Experiment : 5



RStudio interface showing a script with the following code:

```
1 a = c("Short", "Medium", "Tall", "Medium", "Tall", "Short")
2 b = factor(a)
3 print(b)
4
5 set.seed(10)
6 c = sample(LETTERS[1:5], 8, replace = TRUE)
7 d = factor(c)
8 print(d)
```

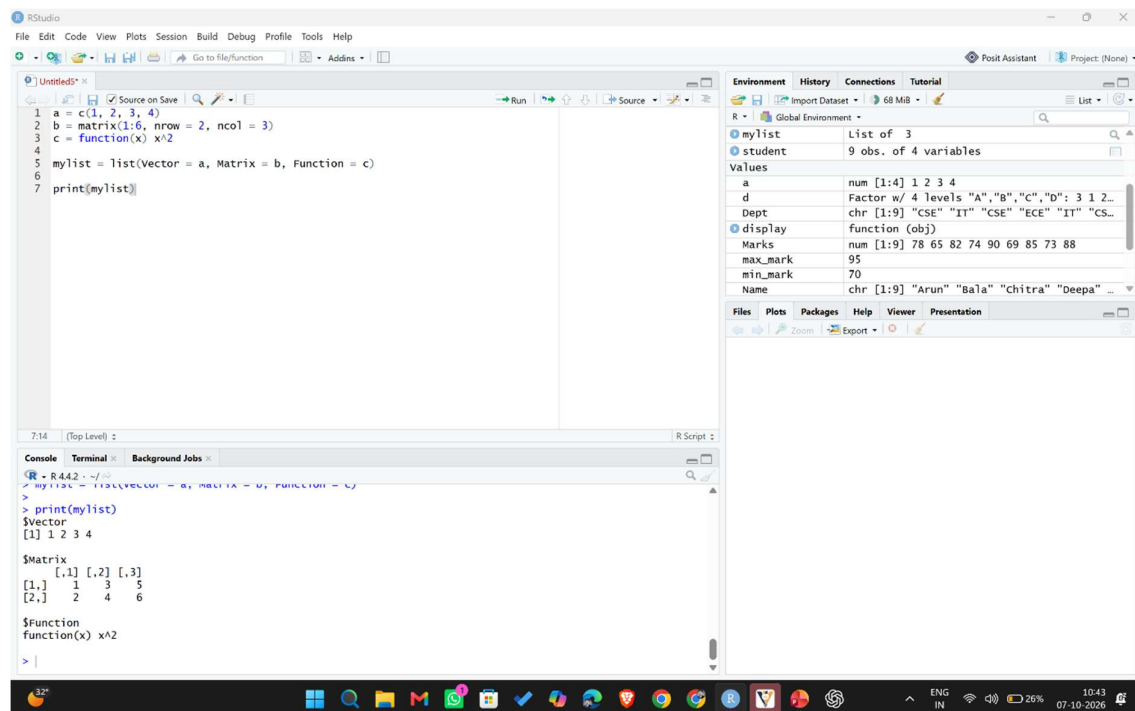
The Environment pane on the right shows the following objects:

Object	Class	Value
a	chr [1:6]	"Short" "Medium" "Tall" "Medium" "Tall" "Short"
b	Factor w/ 3 levels	"Medium", "Short", ...
c	chr [1:8]	"c" "A" "b" "d" "c" "b" "b" "b"
d	Factor w/ 4 levels	"A", "b", "c", "d": 3 1 2 ...
Dept	chr [1:9]	"CSE" "IT" "CSE" "ECE" "IT" "CS..."
display	function (obj)	
Marks	num [1:9]	78 65 82 74 90 69 85 73 88
max_mark	95	
min_mark	70	
Name	chr [1:9]	"Arun" "Bala" "Chitra" "Deepa" ...
overall_average	81.7666666666667	

The Console shows the output of the script:

```
> a = c("Short", "Medium", "Tall", "Medium", "Tall", "Short")
> b = factor(a)
> print(b)
[1] Short Medium Tall Medium Tall Short
Levels: Medium Short Tall
>
> set.seed(10)
> c = sample(LETTERS[1:5], 8, replace = TRUE)
> d = factor(c)
> print(d)
[1] C A B D C B B B
Levels: A B C D
```

Experiment : 6



RStudio interface showing a script with the following code:

```
1 a = c(1, 2, 3, 4)
2 b = matrix(1:6, nrow = 2, ncol = 3)
3 c = function(x) x^2
4
5 mylist = list(Vector = a, Matrix = b, Function = c)
6
7 print(mylist)
```

The Environment pane on the right shows the following objects:

Object	Class	Value
mylist	List of 3	
student	9 obs. of 4 variables	
a	num [1:4]	1 2 3 4
d	Factor w/ 4 levels	"A", "b", "c", "d": 3 1 2 ...
Dept	chr [1:9]	"CSE" "IT" "CSE" "ECE" "IT" "CS..."
display	function (obj)	
Marks	num [1:9]	78 65 82 74 90 69 85 73 88
max_mark	95	
min_mark	70	
Name	chr [1:9]	"Arun" "Bala" "Chitra" "Deepa" ...

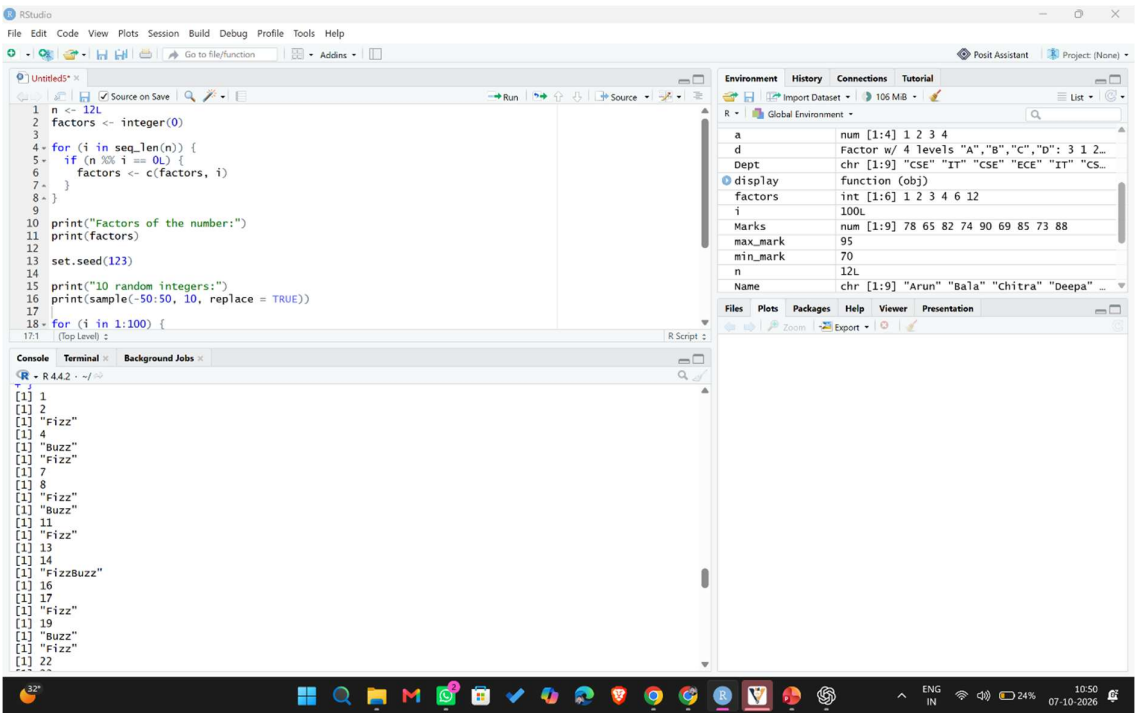
The Console shows the output of the script:

```
> mylist = list(Vector = a, Matrix = b, Function = c)
> print(mylist)
$vector
[1] 1 2 3 4

$Matrix
      [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

$function
function(x) x^2
```

Experiment : 7



```
1 n <- 12L
2 factors <- integer(0)
3
4 for (i in seq_len(n)) {
5   if (n % 30 == 0L) {
6     factors <- c(factors, i)
7   }
8 }
9
10 print("Factors of the number:")
11 print(factors)
12
13 set.seed(123)
14
15 print("10 random integers:")
16 print(sample(-50:50, 10, replace = TRUE))
17
18 for (i in 1:100) {
19   # FizzBuzz logic
20 }
```

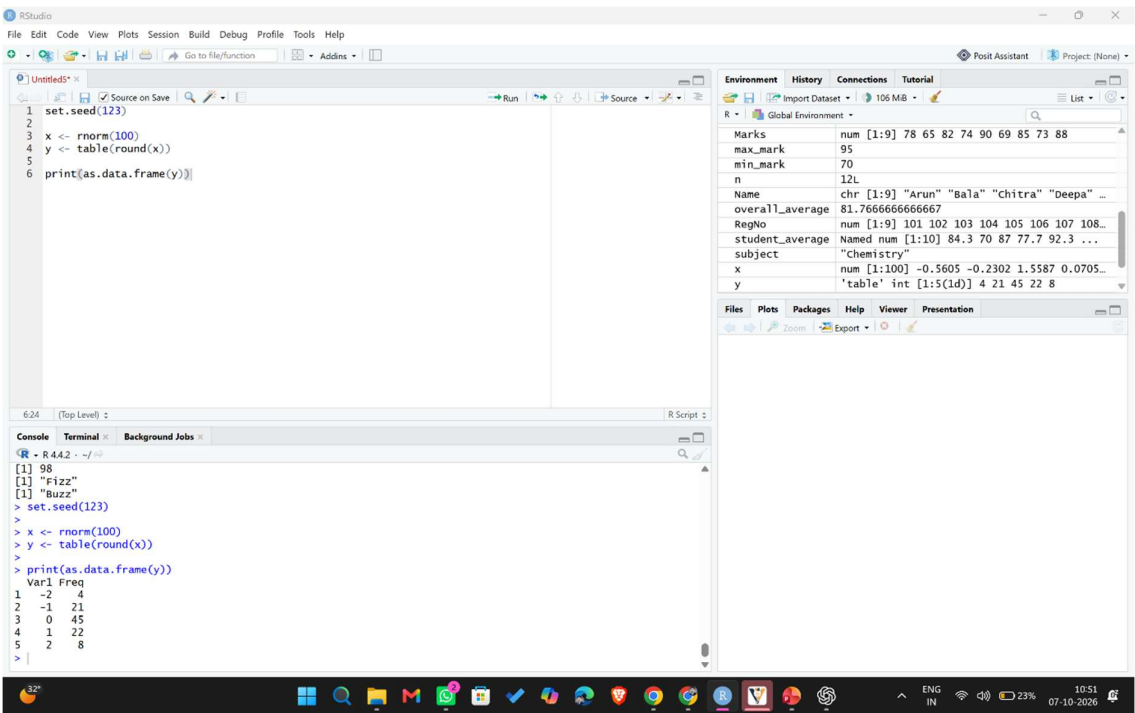
Console Output:

```
[1] 1
[1] 2
[1] "Fizz"
[1] 4
[1] "Buzz"
[1] "Fizz"
[1] 7
[1] 8
[1] "Fizz"
[1] "Buzz"
[1] 11
[1] "Fizz"
[1] 13
[1] 14
[1] "FizzBuzz"
[1] 16
[1] 17
[1] "Fizz"
[1] 19
[1] "Buzz"
[1] "Fizz"
[1] 22
```

Environment:

Object	Class	Value
a	num	[1:4] 1 2 3 4
d	Factor w/ 4 levels	"A","B","C","D": 3 1 2...
Dept	chr	[1:9] "CSE" "IT" "CSE" "ECE" "IT" "CS..."
display	function (obj)	
factors	int	[1:6] 1 2 3 4 6 12
i	100L	
Marks	num	[1:9] 78 65 82 74 90 69 85 73 88
max_mark	95	
min_mark	70	
n	12L	
Name	chr	[1:9] "Arun" "Bala" "Chitra" "Deepa" ...

Experiment : 8



```
1 set.seed(123)
2
3 x <- rnorm(100)
4 y <- table(round(x))
5
6 print(as.data.frame(y))
```

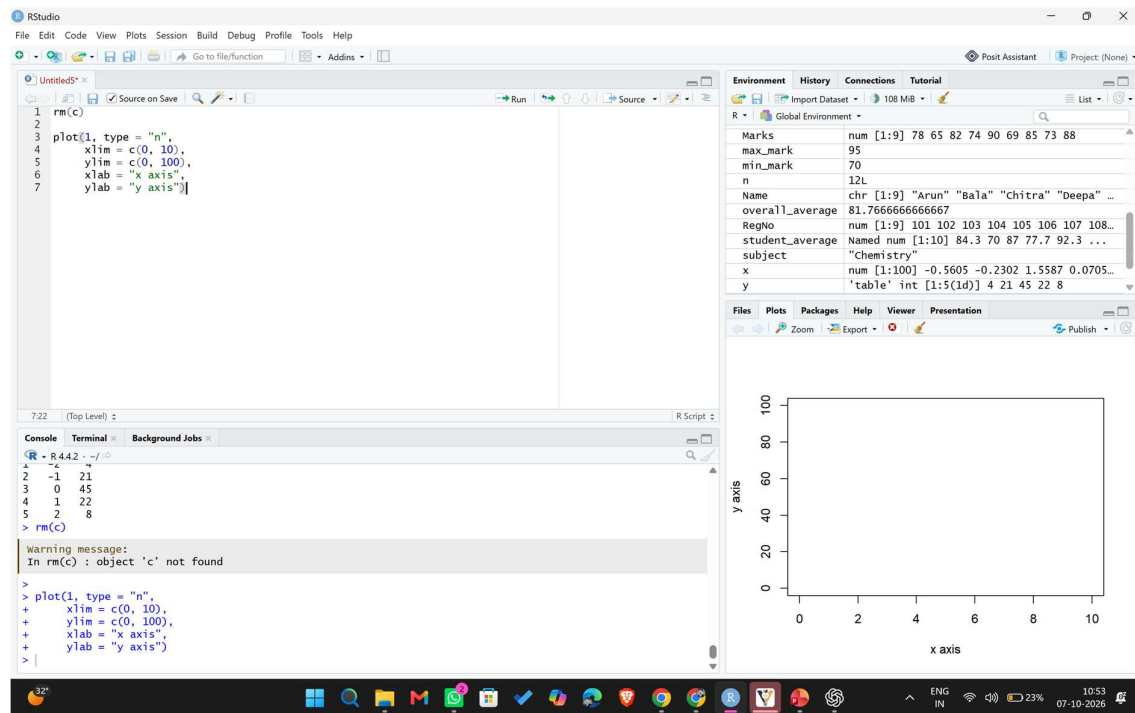
Console Output:

```
[1] 98
[1] "Fizz"
[1] "Buzz"
> set.seed(123)
>
> x <- rnorm(100)
> y <- table(round(x))
> print(as.data.frame(y))
Var1 Freq
1 -2 4
2 -1 21
3 0 45
4 1 22
5 2 8
```

Environment:

Object	Class	Value
Marks	num	[1:9] 78 65 82 74 90 69 85 73 88
max_mark	95	
min_mark	70	
n	12L	
Name	chr	[1:9] "Arun" "Bala" "Chitra" "Deepa" ...
overall_average	81.7666666666667	
RegNo	num	[1:9] 101 102 103 104 105 106 107 108...
student_average	Named num	[1:10] 84.3 70 87 77.7 92.3 ...
subject	"chemistry"	
x	num	[1:100] -0.5605 -0.2302 1.5587 0.0705...
y	'table' int	[1:5(1d)] 4 21 45 22 8

Experiment : 9



Experiment : 10

